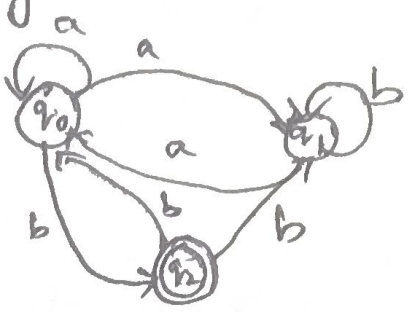
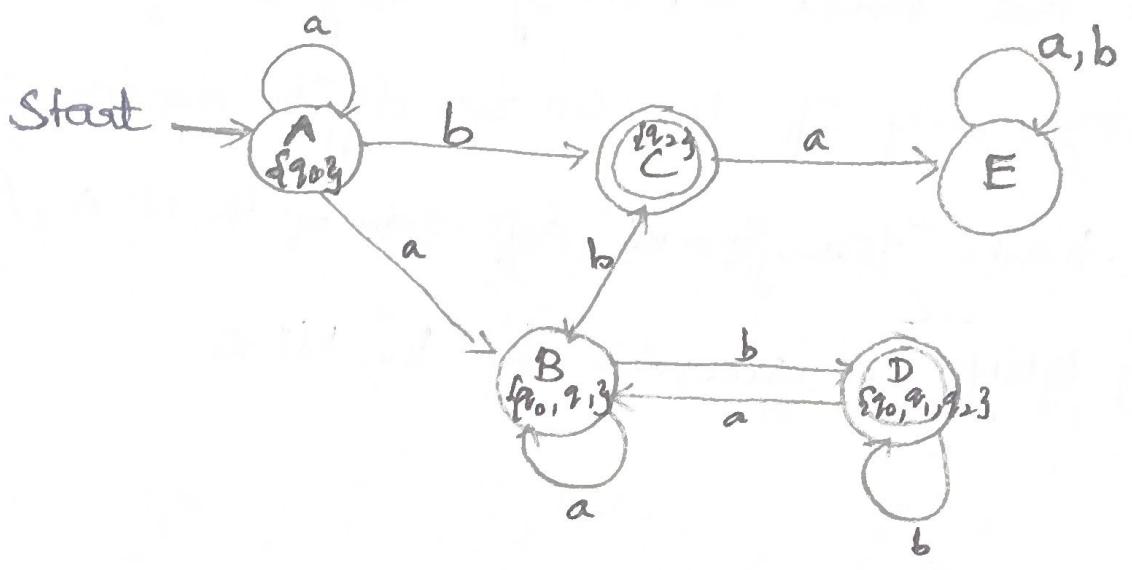
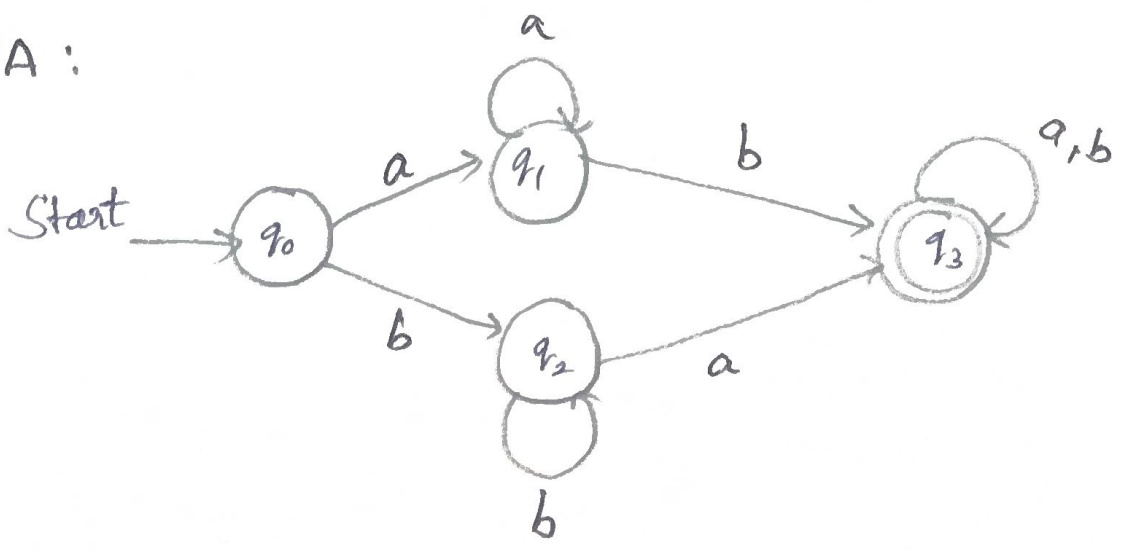
1) Consider the NFA given below which accepts some Language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L.



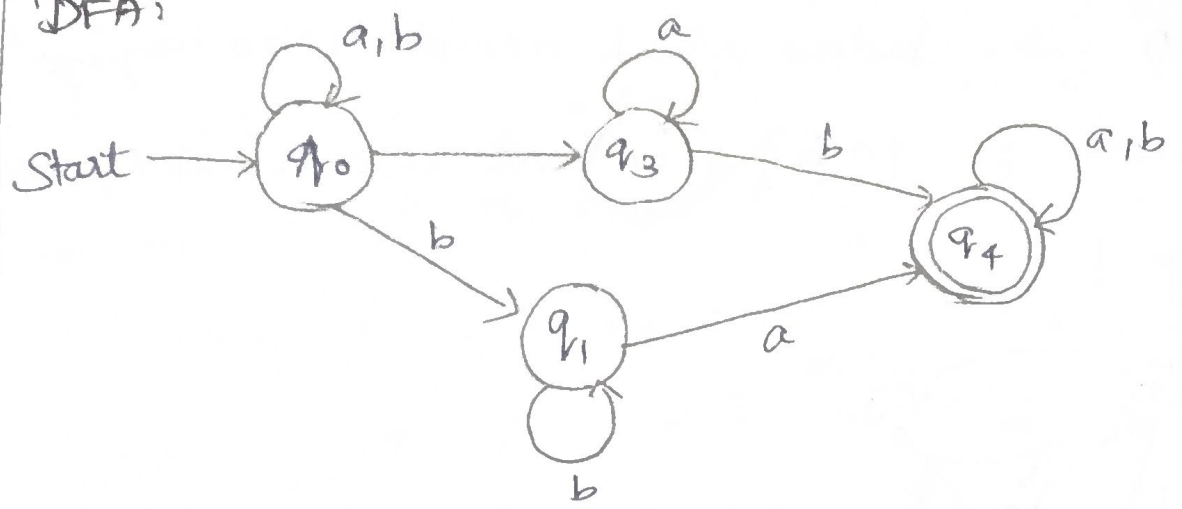


2) Let L be the language of words of length atleast 2 whose last two letters are different from each other. For example abaab is L but not bb, Construct an NFA for L, Then draw the equivalent DFA.

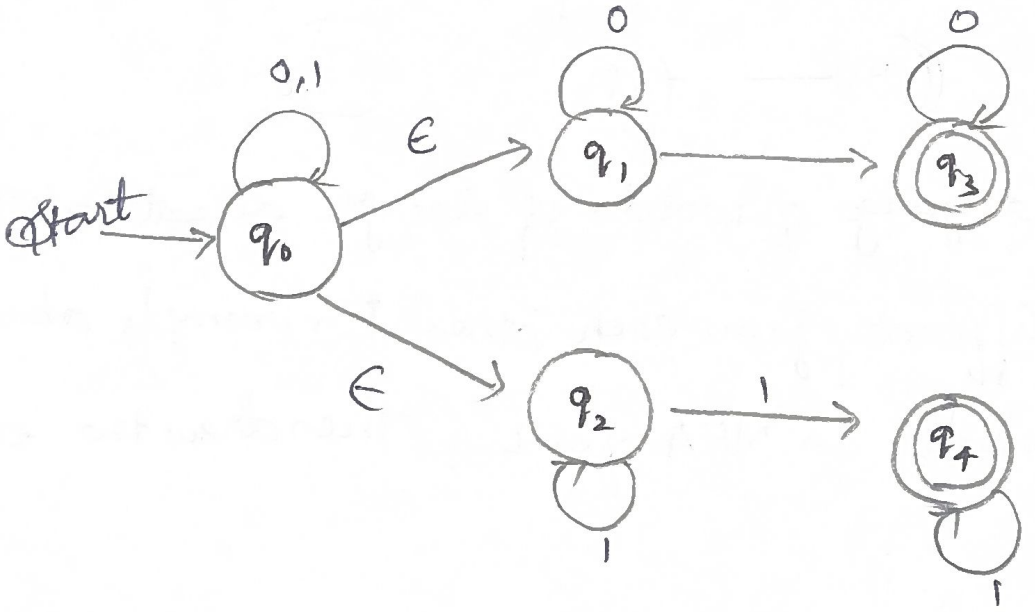
NFA:



DFA:



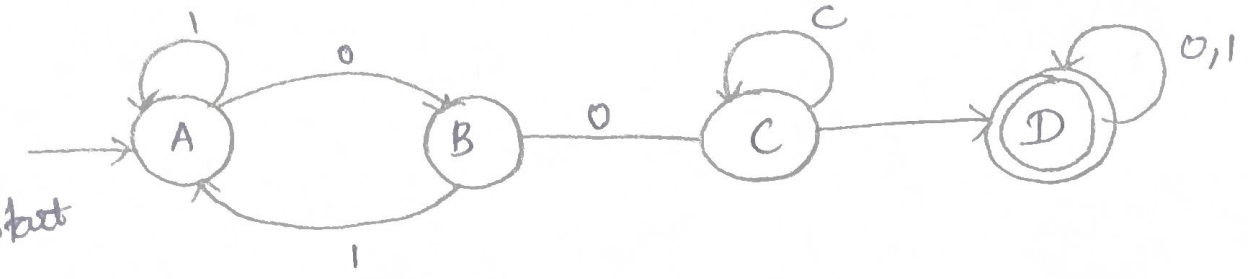
3) Design a NFA that takes a binary string as input and accepts the strings only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that string 01011 is accepted by the NFA.



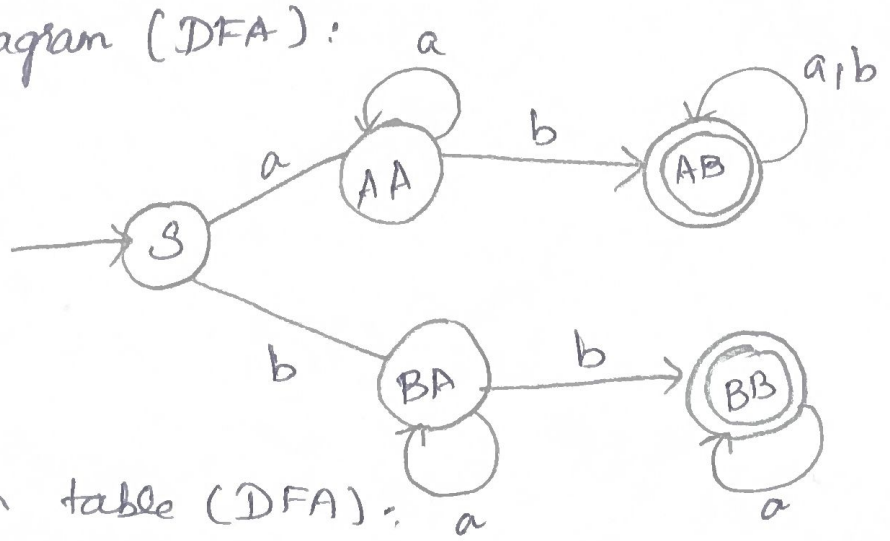
4) Consider the NFA given below which appears (accepts) the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language.

NFA to DFA for strings containing substring 001

DFA diagram:



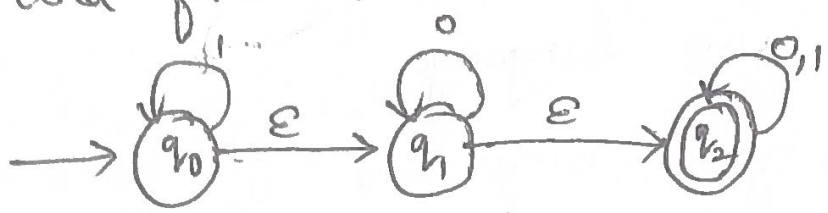
State Diagram (DFA):



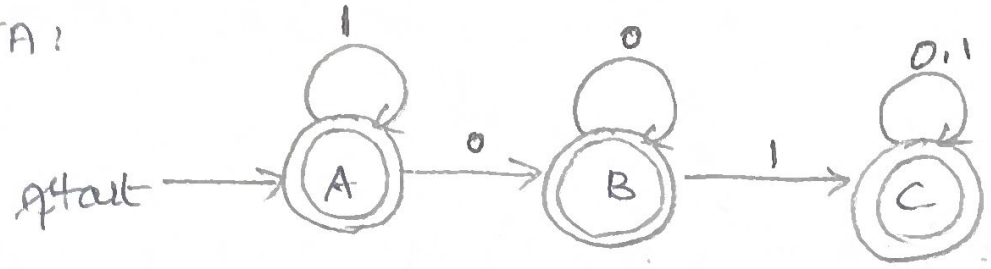
Transition table (DFA):

States	a	b
S	A	B
A	AA	AB
B	BA	BB
AA	AA	AB
AB	BA	BB
BA	AA	AB
BB	BA	BB

5) A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states.



DFA:



Transition Table:

State	0	1
A	B	A
B	B	C
C	C	C